

The Hierarchical Sensor-Control

Chapter II — On Levels, Fan-In, and Correction Without Broadcast

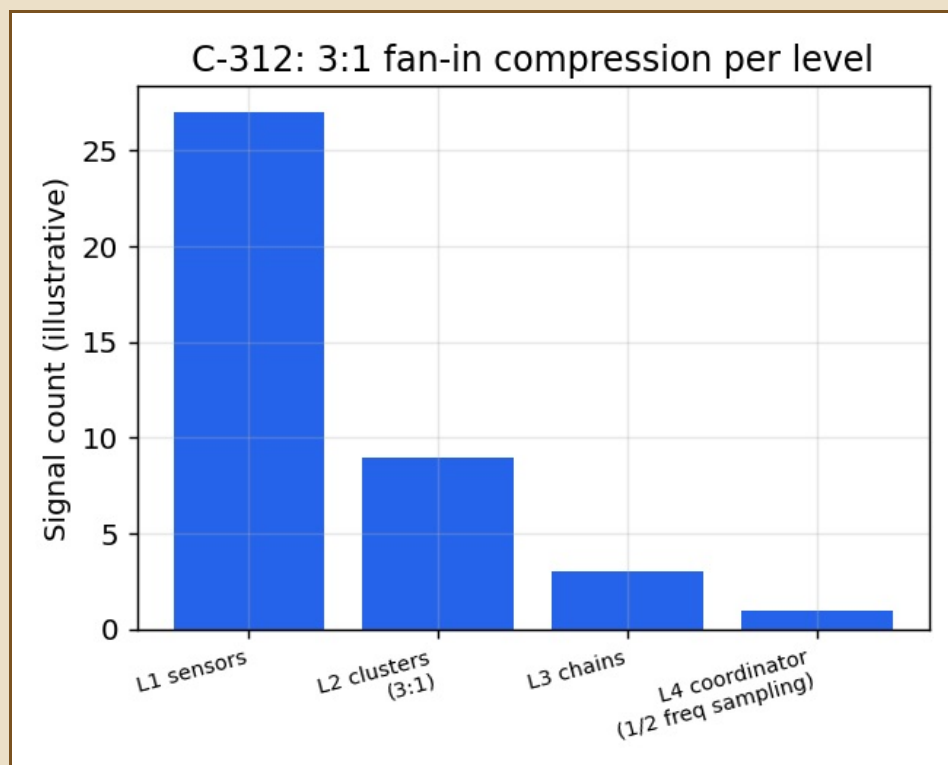
A body, to move at all, must first agree with itself about who reports to whom. The architecture below is not a claim about where awareness lives — that framing was tried once, and declined, and stays declined. What follows is engineering, plainly stated.

I. Four Levels, Distributed Not Dictatorial

Local sensor units hold their own state and answer to no one moment-to-moment. Clusters compress three such units into one signal. Chains carry that signal onward. A coordinator samples — at half the base signaling frequency — and corrects, but does not dictate.

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LEVEL 1  Local Sensors      (autonomous, no neighbor awareness)
| 3:1 fan-in
LEVEL 2  Local Aggregation (clusters)
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LEVEL 3  Positional Chains
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LEVEL 4  Central Coordinator (samples at half base frequency)
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FIG. 2 — THE FOUR-LEVEL HIERARCHY, 3:1 COMPRESSION AT EVERY STAGE



Signal count compressing 3:1 at each level, sensors to coordinator.

II. The Correction Is Targeted, Not Broadcast

When the coordinator corrects, it addresses only the unit whose signal triggered the

correction — not the whole cluster. Two units that were reporting correctly receive no command at all. This is not generosity. It is bandwidth, and it is also, not incidentally, the same distributed-autonomy principle that governs real spinal motor control.

YELLOW AUDIT

The position-encoding scheme along a chain has no formal definition yet. Command latency through all four levels is undefined. Whether four levels is the right number — rather than three, or five — remains untested.